



Project Preliminary Design

SSW 423: Software Engineering Design VII

FindIt – Group 2

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Team Introductions

- Ryan Davis
 - 4/4 Software Engineering
- Roy Tas
 - 4/4 Engineering Management
- Cory Vitanza
 - 4/4 Software Engineering



Project Description

- FindIt is a LoRaWAN-based location tracking and data communication system which aims to provide long-range, low-power tracking and real-time monitoring.
- The system targets healthcare and caregiving applications, allowing for caregivers to monitor movement patterns and receive alerts for potential safety concerns.
- It connects our physical trackers to a web platform, making it easier for caregivers to stay aware of location and safety changes.

Agile Process and Preliminary Design

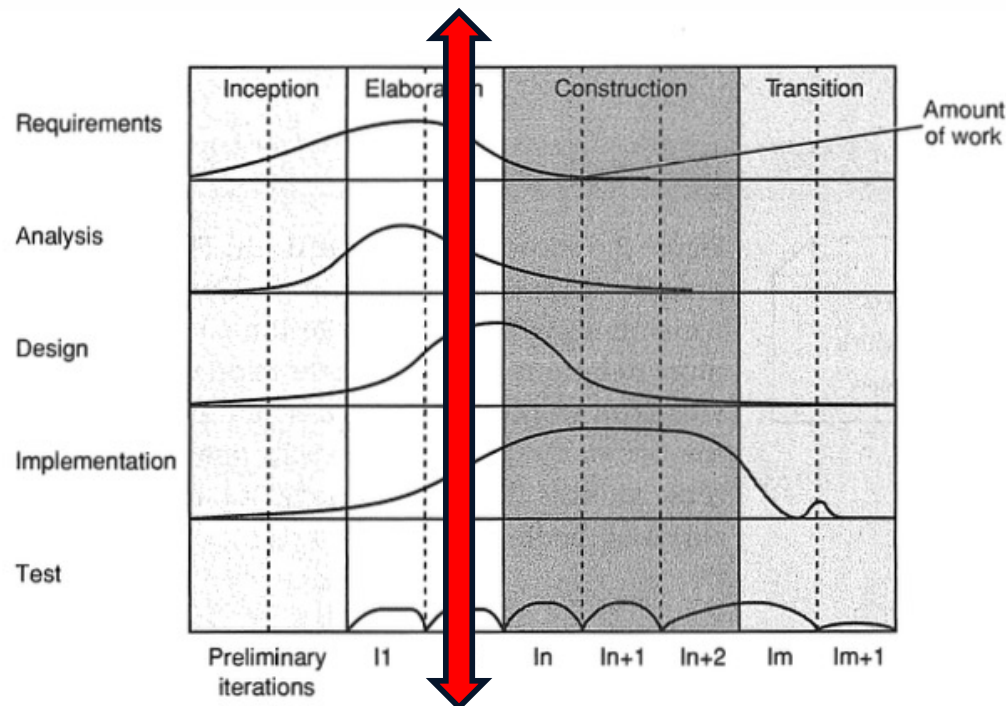


Figure 2.7 Adapted from Figure 1.5 [Jacobson 1] with permission from Addison-Wesley

- Logical View- Class Diagram(s)
- Process View- Activity Diagram
- Implementation View- Component Diagram
- Deployment View- Package Diagram

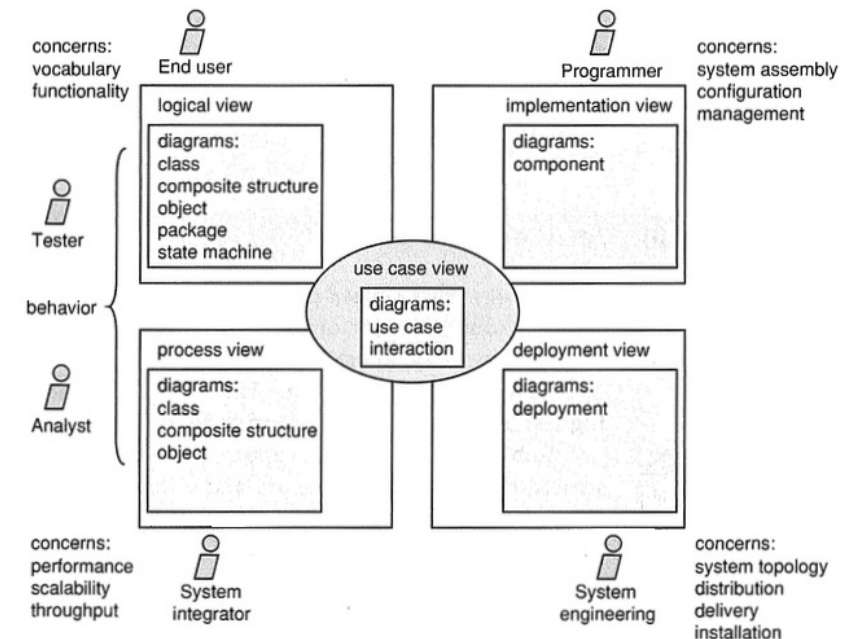


Figure 1.13 Adapted from Figure 5.1 [Kruchten 1] with permission from Addison-Wesley



Remind Audience about Development Plan

- **Milestone 1: Project Setup & Requirements Finalization**
 - **Deliverables:** Project repository initialized, Kanban board configured, requirements document finalized, hardware/software resources received and confirmed.
- **Milestone 2: Core System Architecture & Prototype Communication**
 - **Deliverables:** Draft system architecture, initial Arduino board programming for basic LoRaWAN communication, mock frontend interface, database schema in Supabase.
- **Milestone 3: Integration of Hardware and Software Components**
 - **Deliverables:** Hardware successfully transmitting data to backend, frontend connected to database, Docker setup for local deployment, integration testing initiated.
- **Milestone 4: Feature Implementation & System Testing**
 - **Deliverables:** Full set of planned features implemented, CI/CD pipeline operational, unit and integration tests passing, preliminary user documentation drafted.
- **Milestone 5: Final Review & Delivery**
 - **Deliverables:** Final acceptance testing completed, documentation finalized, project demonstration delivered, final repository and deployment package prepared for submission.



Remind Audience of Major User Requirements

FR1: The system shall support geofencing with alerts when a tracker exits predefined boundaries.	MustHave	ucGeofenceAlert
FR2: The system shall provide real-time location updates via the caregiver mobile app.	MustHave	ucGeofenceAlert
FR3: The system shall support simultaneous monitoring of multiple trackers.	MustHave	ucMultiResidentMonitoring
FR5: The device shall monitor remaining battery capacity and send low-power alerts.	MustHave	ucBatteryStatusAlert
FR7: The system shall include a button that triggers an emergency alert notification to caregivers.	ShouldHave	ucEmergencyAlert
CR1: The system shall operate without subscription fees by using community LoRaWAN gateways.	MustHave	ucCostEfficientDeployment
QR2: The ML model shall maintain accuracy in detecting anomalies without degrading battery life.	CouldHave	ucPatternDetection
NFR3: The caregiver app shall display alerts within two screen interactions.	ShouldHave	Usability
NFR4: The system shall be modular, allowing firmware or API updates independently.	ShouldHave	Maintainability



User Personas

Maria Davis

- **Age:** 42 years old
- **Family Status:** Full-time caregiver for her 80-year-old mother (early dementia)
- **Location:** Brooklyn, NY
- **Occupation:** Part-time dental assistant
- **Background:** Moved her mother in last year after wandering incidents
- **Core Challenge:** Balancing work, kids, and caregiving; high morning anxiety
- **Required Solution Features:**
 - Simple and dependable device
 - Immediate geo-fence notifications
 - Reduced need for continuous supervision

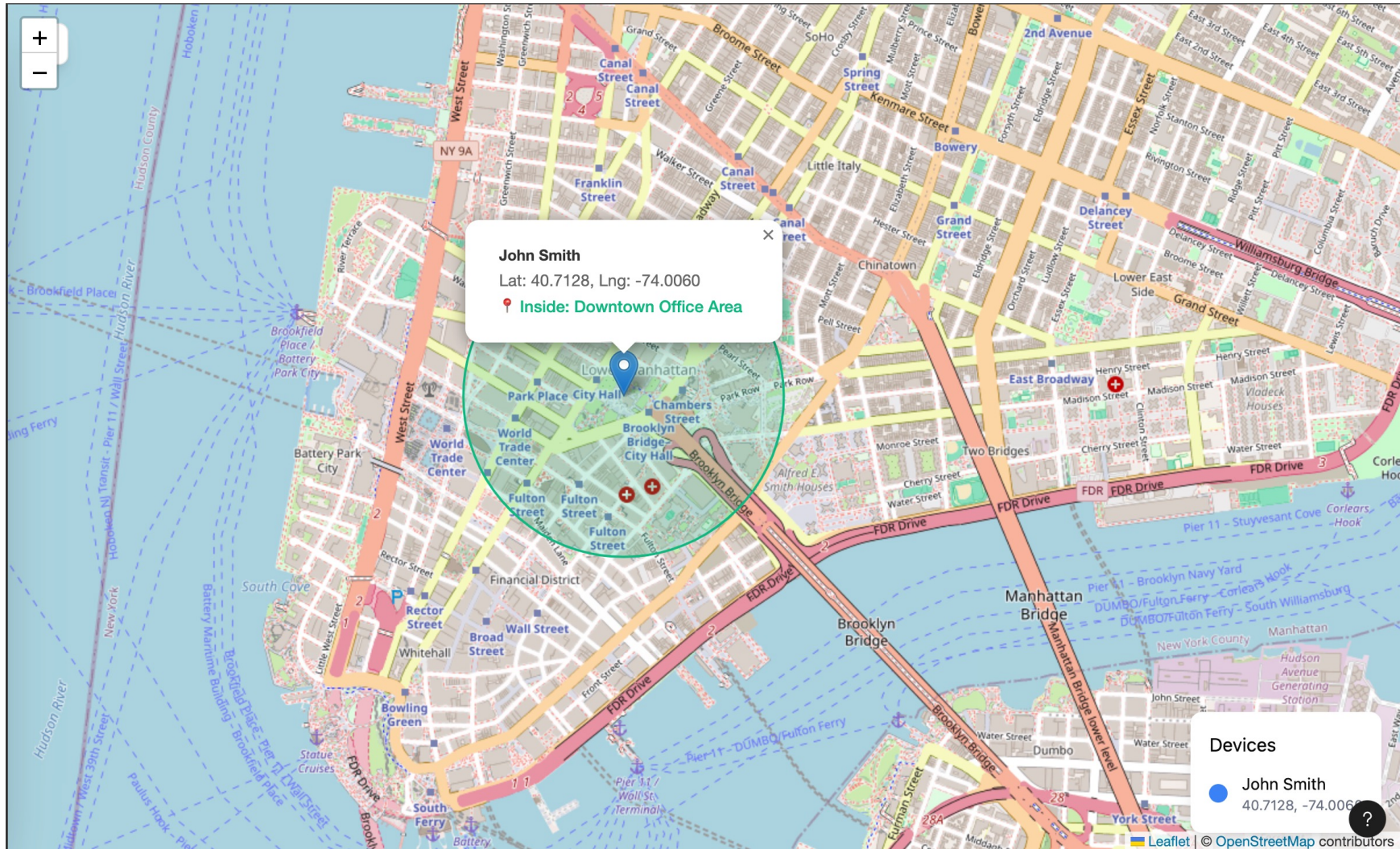
John Smith

- **Age:** 10 years old
- **Condition:** Diagnosed with autism
- **Location:** Paramus, NJ
- **Wandering Triggers:** Stressful or crowded environments (malls, airports)
- **Personality:** Smart, curious, active; enjoys technology
- **Device Preferences:**
 - Prefers small wearable devices
 - Dislikes bulky devices
- **Required Solution Features:**
 - Real-time location updates for parents
 - Quick distance + geofence alerts
 - Simple UI for school staff

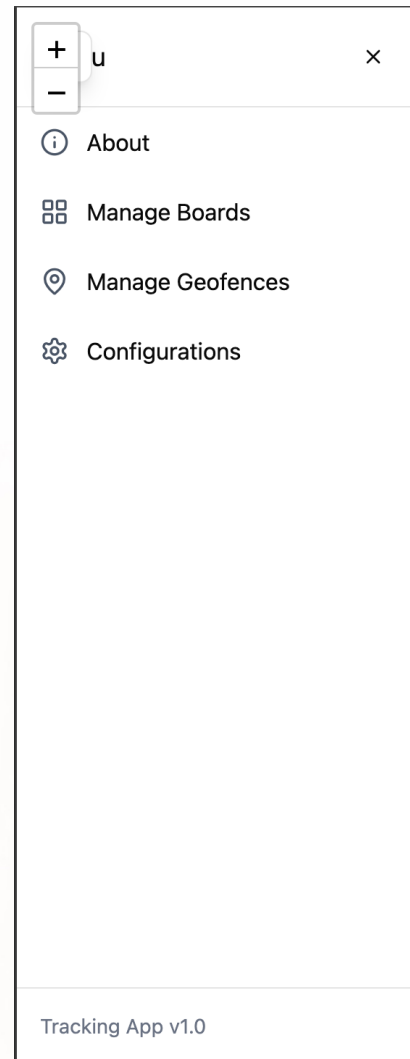
Brian Cleary

- **Age:** 68 years old
- **Status:** Retired mechanic living alone
- **Location:** Weehawken, NJ
- **Health Concerns:** Dizziness; fall risk; forgets to charge devices
- **Routine:** Morning walks by the Hudson River
- **Device Preferences:**
 - Prefers extremely simple or automatic devices
 - Avoids smartphones
- **Required Solution Features:**
 - Long battery life
 - Automatic fall detection
 - Passive well-being updates for family

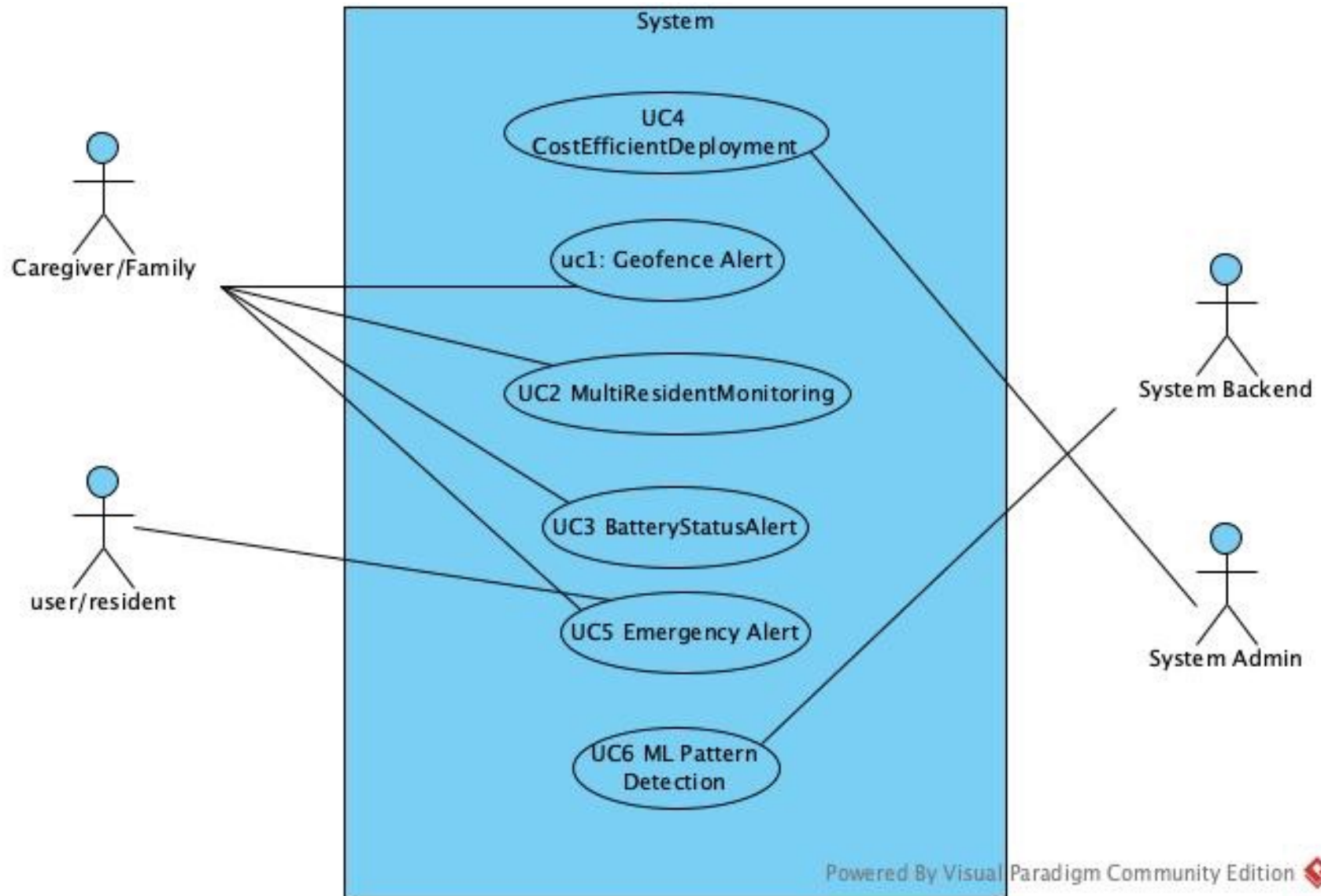
UI Prototyping



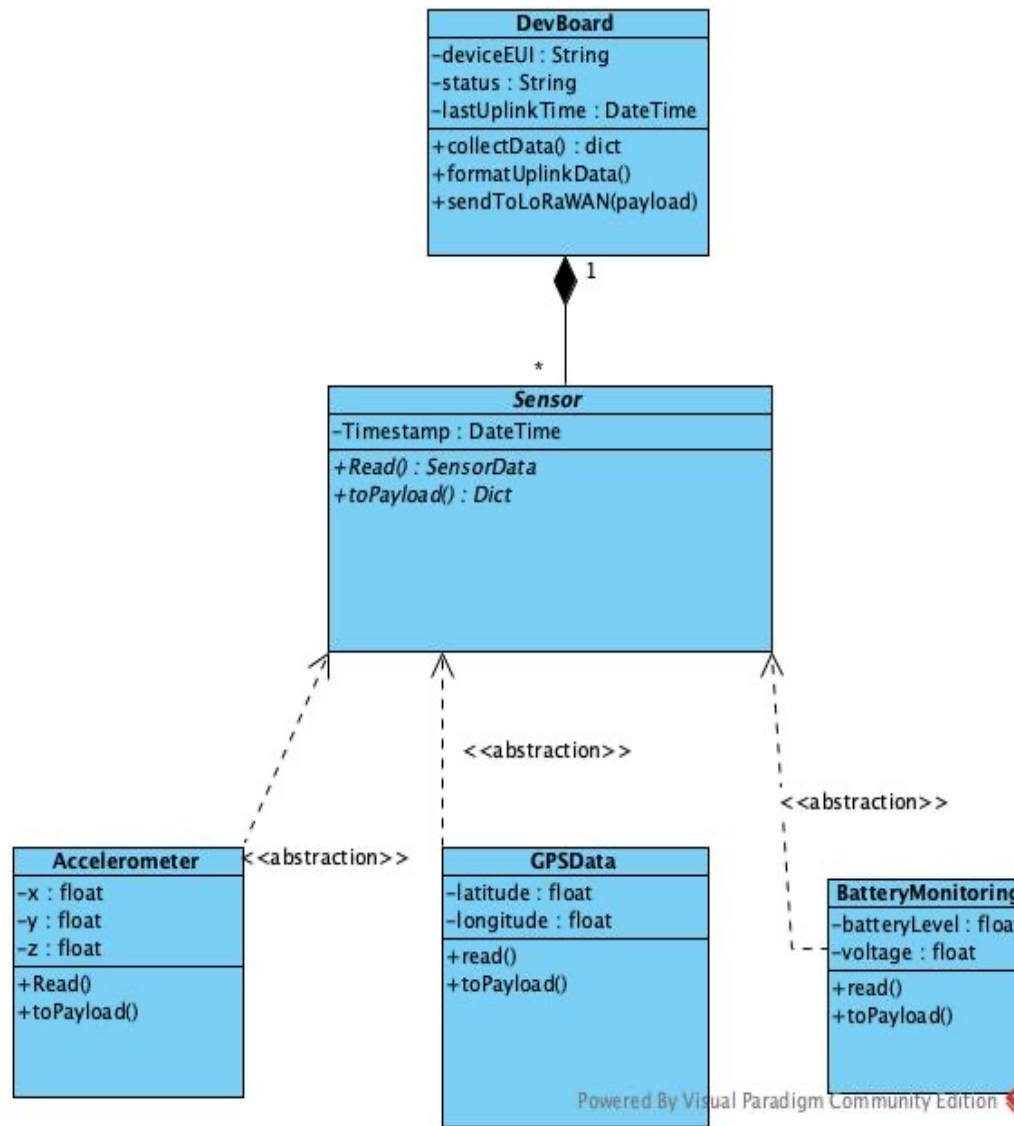
UI Prototyping



Use Case Modeling



Logical View (Class Diagram- Firmware)

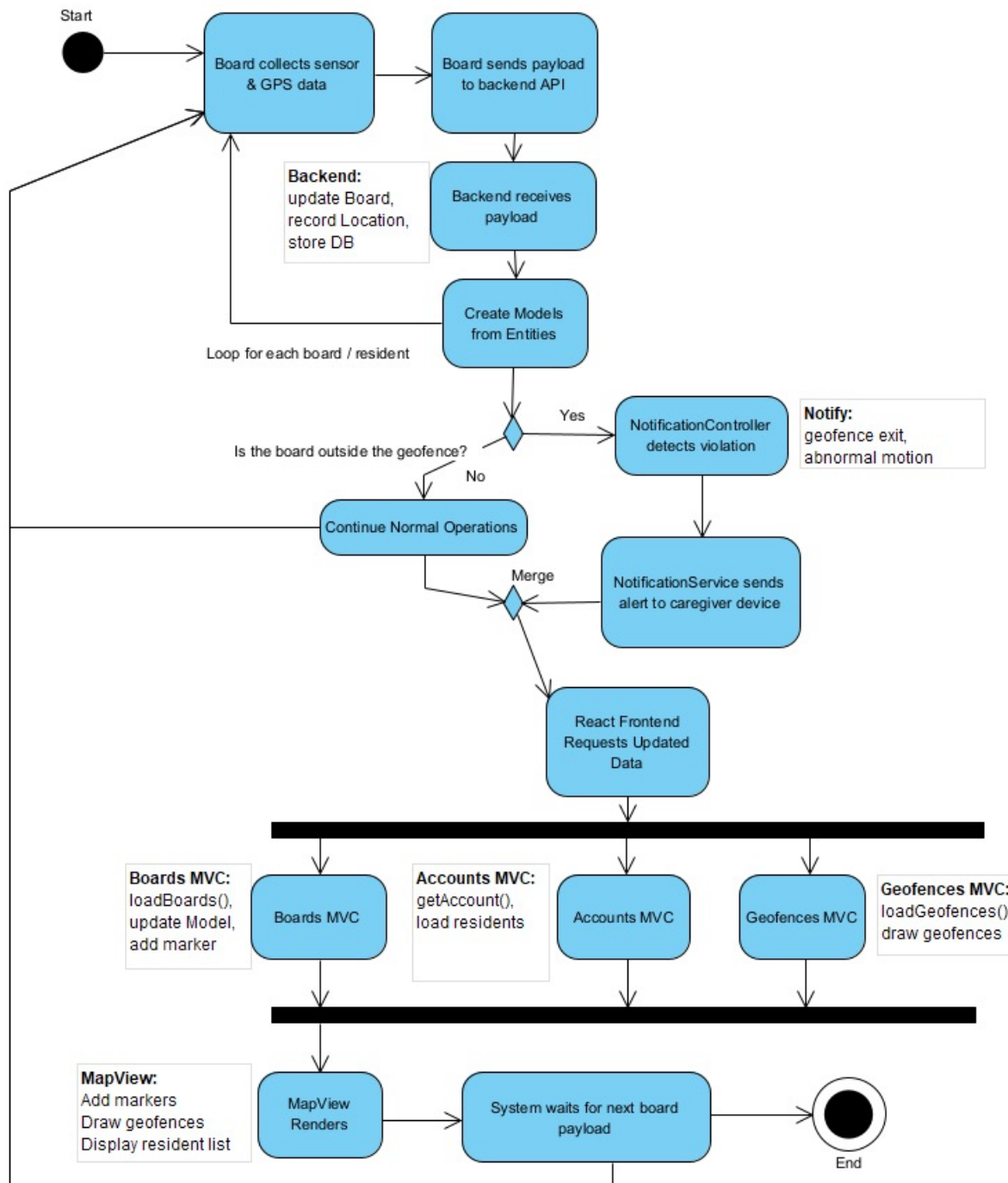






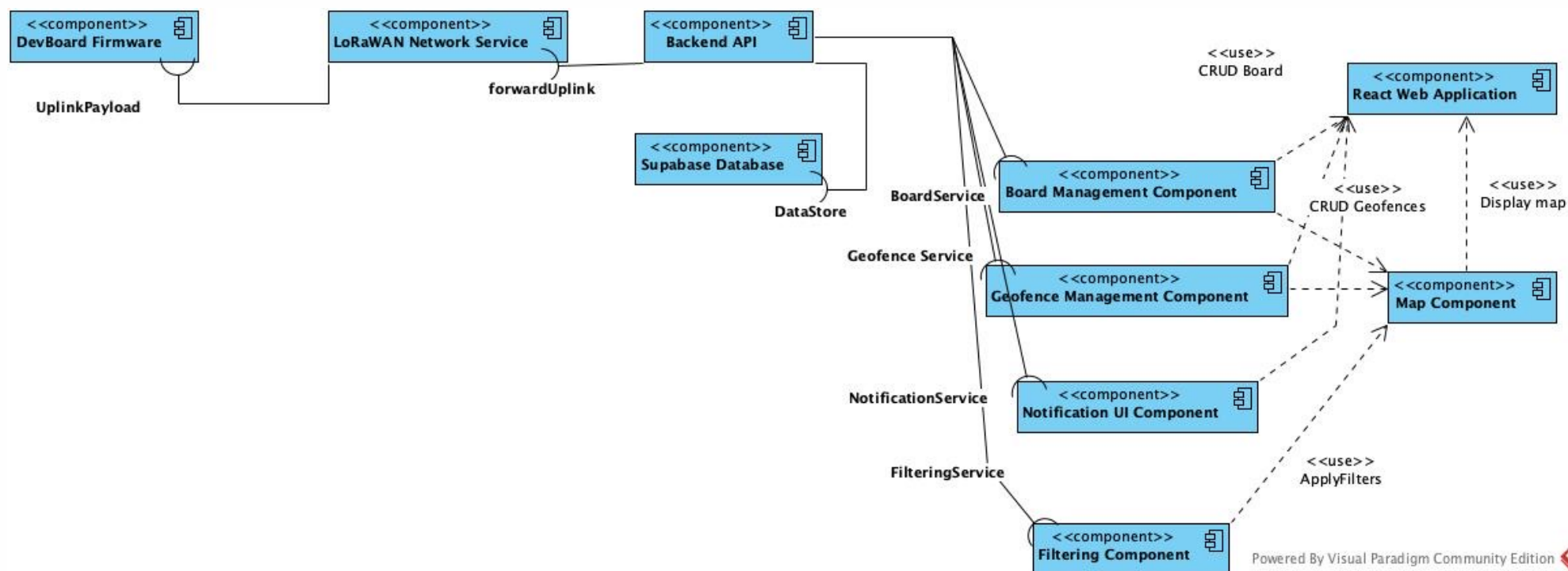
Process View

Activity Diagram



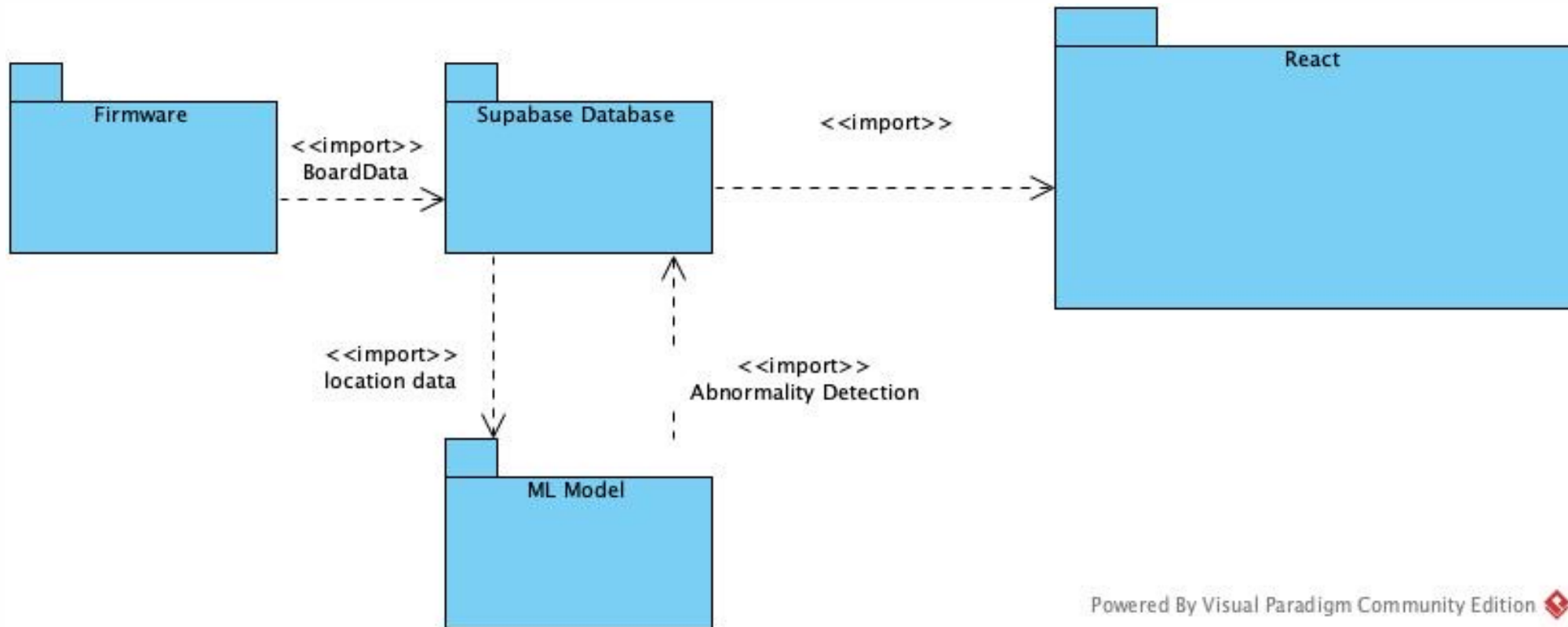


Implementation View (Component Diagram)





Deployment View – Package Diagram





Remind Audience of Software Versioning

Major

- Fundamental Changes to structural components (User interface, Firmware Updates, Full Software release).

Minor

- Feature additions to dashboard, database schema updates, documentation expansions, or design updates

Changelog

- Bug fixes, user interface improvements, minor optimizations, or weekly reports

Version 0.3.11 (Manual)



PDF Manual Demo

